

B1 Algebra — Worksheet

IGCSE Mathematics 0580 | Practice Problems

B1_Q1: Algebraic Expressions — Problems

F

Foundation

[1 mark]

Q1. Simplify: $5x + 3y - 2x + 7y$

F

Foundation

[1 mark]

Q2. Expand: $4(2x + 3)$

F

Foundation

[1 mark]

Q3. Simplify: $7a + 3b - 2a - 5b + 4$

F

Foundation

[1 mark]

Q4. Expand and simplify: $3(2x - 1) + 4$

C

Core

[1 mark]

Q5. Expand and simplify: $2(3x + 4) + 3(2x - 5)$

C Core [1 mark]

Q6. Expand: $(x + 3)(x + 7)$

C Core [1 mark]

Q7. Simplify: $5(2x - 1) - 3(x + 4) + 2x$

C Core [1 mark]

Q8. Expand: $(x + 5)^2$

H Challenge [1 mark]

Q9. Expand and simplify: $(2x + 1)(3x - 4)$

H Challenge [1 mark]

Q10. Expand: $(2x + 3)(2x - 3)$

H Challenge [1 mark]

Q11. Simplify: $(x + 2)^2 - (x - 3)(x + 3)$

H

Challenge

[1 mark]

Q12. Expand: $(x + 4)(x^2 - 2x + 3)$

X

Extension

[1 mark]

Q13. A cuboid has dimensions $(x+2)$ cm, $(x+1)$ cm, and $(x-1)$ cm. Write an expression for its volume and simplify. Hence find the volume when $x = 5$.

X

Extension

[1 mark]

Q14. Prove that the product of two consecutive odd numbers is always one less than a multiple of 4.

X

Extension

[1 mark]

Q15. Simplify $\frac{x^2 + 4x + 3}{x + 1} + \frac{x^2 - 9}{x - 3}$

X

Extension

[1 mark]

Q16. If $(ax + b)(cx + d) = 6x^2 + 11x - 10$ for all x , and a, b, c, d are integers, find all possible sets of values for a, b, c, d .

B1_Q2: Factorisation — Problems

F

Foundation

[1 mark]

Q17. Factorise fully: $6x + 9$

F

Foundation

[1 mark]

Q18. Factorise: $12x^2 - 8x$

F

Foundation

[1 mark]

Q19. Factorise: $x^2 - 16$

F

Foundation

[1 mark]

Q20. Factorise: $x^2 + 5x + 6$

C

Core

[1 mark]

Q21. Factorise fully: $3x^2 - 27$

C

Core

[1 mark]

Q22. Factorise: $x^2 - 8x + 15$

C Core [1 mark]

Q23. Factorise: $x^2 - x - 12$

C Core [1 mark]

Q24. Factorise: $9x^2 - 4y^2$

H Challenge [1 mark]

Q25. Factorise: $2x^2 + 5x + 2$

H Challenge [1 mark]

Q26. Factorise: $3x^2 - 10x - 8$

H Challenge [1 mark]

Q27. Factorise fully: $6x^2 - 13x + 6$

H

Challenge

[1 mark]

Q28. Factorise: $6x^2 + 5x - 6$ **X**

Extension

[1 mark]

Q29. Factorise fully: $(x^2 + 3x)^2 - 14(x^2 + 3x) + 40$ **X**

Extension

[1 mark]

Q30. Simplify by factorising: $\frac{x^2 - 4}{x^2 - 2x - 8}$ **X**

Extension

[1 mark]

Q31. Prove that $n^3 - n$ is always divisible by 6 for any integer n .**X**

Extension

[1 mark]

Q32. Factorise: $x^3 - 3x^2 - 4x + 12$

B1_Q3: Linear Equations — Problems

F Foundation [1 mark]

Q33. Solve: $x + 7 = 12$

F Foundation [1 mark]

Q34. Solve: $3x = 18$

F Foundation [1 mark]

Q35. Solve: $2x + 5 = 13$

F Foundation [1 mark]

Q36. Solve: $x - 3 > 7$

C Core [1 mark]

Q37. Solve: $5x - 3 = 2x + 9$

C Core [1 mark]

Q38. Solve: $2(3x - 1) = 16$

C Core [1 mark]

Q39. Solve: $3x - 7 \leq 11$

C Core [1 mark]

Q40. Solve: $7 - 2x = 3x + 22$

H Challenge [1 mark]

Q41. Solve: $3(2x - 1) = 4(x + 2)$

H Challenge [1 mark]

Q42. Solve: $12 - 3x \geq 5x + 4$

H Challenge [1 mark]

Q43. Solve: $\frac{x}{2} + \frac{x}{3} = 10$

H Challenge [1 mark]

Q44. Solve: $\frac{2x - 1}{3} = \frac{x + 4}{2}$

X Extension [1 mark]

Q45. A rectangle is 3 cm longer than it is wide. Its perimeter is 34 cm. Find its length and width.

X Extension [1 mark]

Q46. Solve: $-5 < 2x + 3 \leq 9$

X Extension [1 mark]

Q47. Solve: $\frac{2x + 1}{3} - \frac{x - 2}{4} = \frac{5}{6}$

X Extension [1 mark]

Q48. Mary is three times as old as her son. In 12 years, Mary will be twice as old as her son. How old is Mary now?

B1_Q4: Simultaneous Equations — Problems**F** Foundation [1 mark]Q49. Solve: $x + y = 7$ and $x - y = 3$

F Foundation [1 mark]Q50. Solve: $x + y = 10$ and $2x + y = 16$

F Foundation [1 mark]Q51. Solve: $y = 2x$ and $x + y = 9$

F Foundation [1 mark]Q52. Solve: $2x + y = 12$ and $x - y = 3$

C Core [1 mark]Q53. Solve: $3x + 2y = 14$ and $5x + 2y = 22$

C Core [1 mark]Q54. Solve: $2x + 3y = 12$ and $3x + 2y = 13$

C

Core

[1 mark]

Q55. Solve: $y = 3x - 1$ and $2x + 5y = 29$

C

Core

[1 mark]

Q56. Solve: $2x - 3y = 7$ and $5x + 3y = 28$

H

Challenge

[1 mark]

Q57. Solve: $3x - 4y = 5$ and $2x + 5y = 11$

H

Challenge

[1 mark]

Q58. Solve: $\frac{x}{2} + y = 5$ and $x - \frac{y}{2} = 5$

H

Challenge

[1 mark]

Q59. The sum of two numbers is 28 and their difference is 12. Find the two numbers.

H

Challenge

[1 mark]

Q60. Solve: $4x + 3y = 10$ and $3x - 2y = -1$

X

Extension

[1 mark]

Q61. 3 apples and 4 bananas cost 3.10. 5 apples and 2 bananas cost 3.30. Find the cost of one apple and one banana.

X

Extension

[1 mark]

Q62. Solve: $x + y + z = 6$, $x - y + z = 2$, $x + y - z = 0$

X

Extension

[1 mark]

Q63. Determine the value of k for which the equations $2x + 3y = 7$ and $4x + 6y = k$ have infinitely many solutions.

X

Extension

[1 mark]

Q64. Solve: $y = x^2$ and $x + y = 6$

B1_Q5: Algebraic Fractions — Problems

F

Foundation

[1 mark]

Q65. Simplify: $\frac{3x}{6}$

F

Foundation

[1 mark]

Q66. Simplify: $\frac{4x^2}{2x}$

F

Foundation

[1 mark]

Q67. Multiply: $\frac{2x}{3} \times \frac{6}{x}$

F

Foundation

[1 mark]

Q68. Divide: $\frac{x}{2} \div \frac{x}{4}$

C

Core

[1 mark]

Q69. Simplify: $\frac{x^2 - 9}{x + 3}$

C

Core

[1 mark]

Q70. Add: $\frac{3}{x} + \frac{2}{x}$

C Core [1 mark]

Q71. Subtract: $\frac{5}{x+1} - \frac{2}{x+1}$

C Core [1 mark]

Q72. Simplify: $\frac{2x^2 + 6x}{4x}$

H Challenge [1 mark]

Q73. Add: $\frac{2}{x} + \frac{3}{x+1}$

H Challenge [1 mark]

Q74. Simplify: $\frac{x^2 - x - 6}{x^2 - 4}$

H Challenge [1 mark]

Q75. Subtract: $\frac{3}{x-1} - \frac{2}{x+2}$

H

Challenge

[1 mark]

Q76. Solve: $\frac{3}{x} + 2 = \frac{5}{2x}$

X

Extension

[1 mark]

Q77. Simplify: $\frac{\frac{1}{x+1} - \frac{1}{x-1}}{\frac{1}{x+1} + \frac{1}{x-1}}$

X

Extension

[1 mark]

Q78. Solve: $\frac{2x-1}{x+1} - \frac{x-2}{x-1} = 1$

X

Extension

[1 mark]

Q79. Simplify: $\frac{2x+1}{x+2} - \frac{x-3}{x-1}$

X

Extension

[1 mark]

Q80. A tank can be filled by pipe A in x hours and by pipe B in $(x+3)$ hours. Together they take 2 hours. Find x .

B1_Q6: Sequences — Problems

F

Foundation

[1 mark]

Q81. Find the next two terms of the sequence: 2, 5, 8, 11, \dots

F

Foundation

[1 mark]

Q82. Find the n th term of the sequence: 7, 10, 13, 16, \dots

F

Foundation

[1 mark]

Q83. A sequence has n th term $5n - 2$. Find the 10th term.

F

Foundation

[1 mark]

Q84. The term-to-term rule of a sequence is 'add 4'. The first term is 3. Write the first 5 terms.

C

Core

[1 mark]

Q85. A sequence has n th term $4n - 3$. Which term of the sequence equals 57?

C

Core

[1 mark]

Q86. Find the n th term of the sequence: -1, 3, 7, 11, 15, \dots

C Core [1 mark]

Q87. Find the next term of the quadratic sequence: 1, 4, 9, 16, 25, \dots

C Core [1 mark]

Q88. Find the n th term of the sequence: 11, 7, 3, -1, -5, \dots

H Challenge [1 mark]

Q89. Find the n th term of the sequence: 2, 6, 12, 20, 30, \dots

H Challenge [1 mark]

Q90. Find the n th term of the sequence: 5, 12, 23, 38, 57, \dots

H Challenge [1 mark]

Q91. A sequence has n th term $n^2 + 3n$. Find the value of n for which the term equals 54.

H Challenge [1 mark]

Q92. The n th term of a sequence is $4n^2 - 3n + 1$. Find the 5th term and the first term.

X Extension [1 mark]

Q93. The sum of the first n terms of a sequence is given by $S_n = 2n^2 + 3n$. Find the n th term of the sequence.

X Extension [1 mark]

Q94. A sequence has n th term $an^2 + bn + c$. The first three terms are 2, 9, 22. Find a , b , and c .

X Extension [1 mark]

Q95. Prove that the terms of the sequence $u_n = n^2 + 3n + 1$ are always odd for integer values of n .

X Extension [1 mark]

Q96. A pattern is made from matchsticks forming squares in a row. 1 square needs 4 matches, 2 squares need 7 matches, 3 squares need 10 matches. Find an expression for the number of matches needed for n squares. How many squares can be made from 61 matches?

B1_Q7: Functions and Graphs — Problems

F Foundation [1 mark]

Q97. If $f(x) = 2x + 5$, find $f(3)$.

F Foundation [1 mark]

Q98. If $g(x) = x^2 - 4$, find $g(3)$ and $g(0)$.

F Foundation [1 mark]

Q99. Find the gradient of the line $y = 3x + 2$.

F Foundation [1 mark]

Q100. Find the y-intercept of the line $y = -2x + 7$.

C Core [1 mark]

Q101. Find the gradient of the line passing through (1, 3) and (4, 9).

C Core [1 mark]

Q102. Find the equation of a line with gradient 3 passing through (2, 5).

C

Core

[1 mark]

Q103. If $f(x) = 3x - 2$, find x such that $f(x) = 13$.

C

Core

[1 mark]

Q104. Find the equation of the line through $(0, -3)$ and $(4, 5)$.

H

Challenge

[1 mark]

Q105. Find the gradient of a line perpendicular to $y = 2x + 3$.

H

Challenge

[1 mark]

Q106. Find the equation of the line perpendicular to $y = -3x + 4$ passing through $(2, 1)$.

H

Challenge

[1 mark]

Q107. Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 5)$.

H

Challenge

[1 mark]

Q108. If $f(x) = 2x - 1$ and $g(x) = x^2$, find $f(g(3))$ and $g(f(3))$.

X

Extension

[1 mark]

Q109. If $f(x) = x^2 + 2$ and the domain is $\{-2, -1, 0, 1, 2\}$, find the range.

X

Extension

[1 mark]

Q110. Given $f(x) = \frac{2x+3}{x-1}$, $x \neq 1$, find $f^{-1}(x)$.

X

Extension

[1 mark]

Q111. Find the point of intersection of the lines $y = 3x - 2$ and $2x + y = 8$.

X

Extension

[1 mark]

Q112. A taxi charges a flat fee of 3 plus 2 per kilometre. Write a function $f(x)$ for the cost in dollars for x kilometres. Find the cost for a 12 km journey. How far can you go for 25?